

AMENDMENT NO. 1 SEPTEMBER 1993
TO
IS 9300 (Part 2) : 1989 STATISTICAL MODELS
FOR INDUSTRIAL APPLICATIONS

PART 2 CONTINUOUS MODELS

(First Revision)

(Page 9, clause 9.4) — Insert the following new clause after 9.4:

9.4.1 Example

The following data gives the running time (in hours) of the 40 head boxes. Test whether the data follows the Weibull distribution:

249	259	844	65
303	309	121	32
649	21	146	43
23	104	99	301
130	52	584	21
411	177	583	508
643	281	173	9
180	169	248	42
128	173	524	883
212	547	31	85

The above data is arranged in the form of a frequency table as shown in Table 10.

The next step is to calculate the expected frequencies for each class interval under assumption that the data follows Weibull distribution. For this purpose, one has to estimate the parameters β and λ of the Weibull model. For estimating the parameters, the natural logarithmic values of each of the x_i 's are obtained and thereafter the mean and the standard deviation of these x_i values are calculated.

For this example, mean ($\bar{y}$) = 5.042 1; and

standard deviation (s_y) = 1.16

where, $y_i = \log x_i$.

The estimate of the parameters are then obtained from the following expression:

$$\beta = \frac{\pi}{s_y \sqrt{6}} \text{ and } \lambda = \exp \left[-(\bar{y} + \frac{0.572\,26}{\beta}) \right]$$

For this example, $\beta = 1.106\,1$ and $\lambda = 0.003\,85$

Using the above estimates of the parameters in the frequency distribution of the Weibull model, namely,

$F(x) = 1 - \exp [-(\lambda x)^\beta]$, where, $\beta = 1.106\,1$ and $\lambda = 0.003\,85$, the expected frequency for each class interval may be obtained as given in Table 10.

Goodness of Fit

The calculated value of χ^2 for the example is 1.369 (see Table 11). The tabulated value of χ^2 for 2 degrees of freedom at 5 percent level of significance is 5.99. Since the calculated value is less than the tabulated value, the null hypothesis that the data follows Weibull distribution is accepted.

Table 10 Frequency Table for Running Time of Head Boxes
(Clause 9.4.1)

Class Intervals	Frequency (O_i)	Upper Limit (x)	(λx) ^{β}	F (x)	F (x_i) - F (x_{i-1})	Expected Frequency (E_i)
0-100	12	100	0.347 9	0.293 8	0.293 8	11.75
101-200	10	200	0.748 9	0.527 1	0.233 3	9.33
201-300	5	300	1.171 0	0.690 6	0.163 5	6.54
301-400	3	400	1.612 0	0.800 5	0.109 9	4.40
401-500	3	500	2.064 0	0.873 1	0.072 6	2.90
501-600	2	600	2.525 0	0.919 9	0.046 8	1.87
601-700	2	700	2.994 0	0.949 9	0.030 0	1.20
701-800	2	800	3.470 0	0.968 9	0.019 0	0.76
801-900	1	900	3.953 0	0.980 8	0.011 9	0.48
901 & above	0	—	—	1.000 0	0.019 2	0.77

Table 11 Observed and Expected Frequencies
(*Clause 9.4.1*)

Observed Frequencies (O_i)	Expected Frequencies (E_i)	($O_i - E_i$)	($O_i - E_i$) ² E_i
12	11.75	0.25	0
10	9.33	0.67	0.048
5	6.54	- 1.54	0.363
3 } 6	4.40 } 7.30	- 1.30	0.232
3 }	2.90 }		
2 } 7	1.87 }		
2 }	1.20 }		
2 }	0.76 }	- 1.92	0.726
1 }	0.48 }		
0 }	0.77 }		
Total			1.369

(MSD 3)